

PROJECT PROPOSAL

Module Name & Name of the Module

EBA 5011 Practice Module in Analytics for Business Decisions

Title of Proposal

Proposal for Data-Driven Spatial Priority Modeling for Selection of Public Electric Vehicle (EV) Charging Infrastructure in Singapore

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1. Introduction

Singapore is expanding its public EV charging infrastructure with a target of 60,000 public charging points by 2030. About 80% of residents live in HDB flats with no private garage, so public and communal charging serve as the primary backbone for EV adoption. Currently, Charge Point Operators (CPOs) pick HDB carpark sites reactively, relying on landlord availability, ad-hoc resident requests or unverified demand signals, rather than systematic spatial analysis. This unstructured deployment results in severe spatial utilisation imbalances and increases the risk of localized power grid congestion as the EV fleet scales.

Business Objective: Establish a data driven site selection framework that maximises utilisation of newly deployed chargers, prevents localised power grid congestion, and accelerates capital payback for CPOs and government partners.

Technical Objective: Build a standardised geospatial database merging static HDB carpark data with temporal lot utilisation snapshots, then apply Multi-Criteria Decision Analysis (MCDA) to compute a priority index score across all 2,272 HDB carparks in Singapore.

Potential Approach: An AI-augmented pipeline: AHP-weighted TOPSIS for site ranking, unsupervised clustering for localised demand segmentation, an optional linear-regression calibration layer, and GenAI/LLM integration throughout for insights generation and stakeholder storytelling. Section 4 carries the details.

2. Industry Overview

Policy drivers like the EV Common Charger Grant and Singapore Green Plan 2030 have accelerated EV adoption across Singapore. Public charging is built and managed by CPOs, primarily by SP Mobility, Shell Recharge and Charge+, under commercial and public-private arrangements with HDB and LTA. The structural constraints are sharp. Land and grid capacity are scarce. HDB's housing-dominant profile concentrates demand around public carparks rather than private garages, and every charger incurs substantial upfront CapEx for the CPOs. Site selection likely relied on qualitative judgement rather than spatial analytics. Infrastructure planning is meanwhile moving toward AI-augmented decision support, combining MCDA and spatial statistics with GenAI for automated analysis and stakeholder communication. That is the practice this project applies to Singapore's EV rollout.

3. Business Problem & Objectives

CPOs and government partners select EV charging sites on demand. Utilization ends uneven; clustered deployment carries grid-congestion risk, and capital pays back slower than it needs to, which weakens the case for expanding the network. This project ranks HDB carparks against those factors to guide and benefit both stakeholders.

Business Objective	Outcome Metric	Why Critical
Maximise utilisation	% utilisation of new chargers vs. baseline, 6 months post-launch	Underused chargers depress CPO revenue and slow further rollout.
Prevent grid congestion	# localised congestion incidents near new clusters	Regulatory, remediation, and reputational risk.
Accelerate payback	Average payback period (months) vs. baseline	Improves the investment case for continued expansion.

Technical goals and alignment: The EV Site Priority Index (D2) directly maximizes charger utilization by ranking carparks against localized demand metrics. To mitigate localized grid congestion, the model incorporates an EVCP Saturation criterion and a spatial density proxy that prevents infrastructure over-clustering. Finally, the segmentation model (D3) and interactive dashboard (D4) accelerate capital payback by matching charger types and power capacities to each site's specific demand archetype.

4. Project Design

4.1 Roadmap

Step	Activity	Reference
1	Build a standardised, carpark-level geospatial database from 8 sources	Section 5
2	Compute four MCDA criteria per carpark: EV-Adoption Proxy, Charging Urgency, Commercial Density, EVCP Saturation	Section 4.2
3	Derive an AHP-weighted TOPSIS Priority Index and a K-means / hierarchical carpark segmentation	Section 4.3
4	Run an optional linear-regression calibration layer to sanity-check MCDA weights against observed occupancy	Stretch, Section 5.3
5	Generate a GenAI-augmented dashboard and stakeholder narratives, Insight-to-Impact	D4 (Key Deliverables)
6	Validate by backtesting against known utilisation patterns: national peak average 36%, Tengah 61%, February 2026	D2, D6 (Key Deliverables)

4.2 MCDA Criteria: Formulas & Data-Driven Fixes

Criterion	Formula	Data-driven fix and note
EV-Adoption Proxy (benefit)	$\text{NationalEVShare} = \frac{\text{Total EV registrations}}{\text{Total vehicle registrations (national)}}$ $\text{HouseValueWeight}(P) = \frac{\text{ResalePricePSM}(P)}{\text{NationalMedianResalePricePSM}}$ $\text{ResidentShare}(S) = \frac{\text{ResidentPop}(S)}{\sum \text{ResidentPop}(\text{subzones in } P)}$ $\text{LotShare}(i) = \frac{\text{Lots}(i)}{\sum \text{Lots}(\text{carparks in } S)}$ $\text{EVAdoptionProxy}(i) = \text{NationalEVShare} \times \text{HouseValueWeight}(P) \times \text{ResidentShare}(S) \times \text{LotShare}(i)$	LTA's Fuel-Type Series is a national aggregate with no geographic breakdown, so the national rate is redistributed by HDB resale price, a monthly HDB-specific correlate of ownership used in place of survey-based income, and by subzone population, then disaggregated to each carpark. Resale price is the median \$/sqm, trailing 12 months, by planning area. Resident population is Census 2020 total residents; no HDB-dwelling filter is available. Lots are car lots only, lot_type_code = C
Charging Urgency / Congestion Signal (benefit)	$\text{OccupancyRate}(T) = \frac{\sum \text{occupied EVCP time-slots in town } T}{\sum \text{total available EVCP time-slots in town } T}$ $\text{ChargingUrgency}(i) = \text{OccupancyRate}(T) \times \left[\frac{\text{Lots}(i)}{\sum \text{Lots}(\text{carparks in } T)} \right]$	Kept separate from the EV-Adoption Proxy. Occupancy is undefined in under-served towns and would otherwise conflate demand with scarcity
Commercial Density (benefit)	$\text{CommercialDensity}(i) = \sum \left[\frac{1}{d(b,i)} \right]$ for HDB blocks b within 500m of carpark i where commercial = Y OR market_hawker = Y	No dedicated shopping-centre or Gross Floor Area (GFA) dataset was available. A distance-weighted block count is coarser than a GFA-weighted version, but computable from data on hand
EVCP Saturation (cost)	$\text{EVCPsaturation}(i) = \frac{\text{count}(\text{existing EVCPs at/near carpark } i)}{\text{total car lots at carpark } i}$	A cost criterion, so a lower value raises priority and the index targets coverage gaps rather than reinforcing already-served carparks. Car-lot coverage, Lots(i), is confirmed for 2,003 of 2,272 carparks (88.2%) via the Carpark Availability feed. HDB Carpark Information itself carries no lot-count or usage-type field, confirmed absent rather than merely untested

4.3 Priority Index (TOPSIS) & AHP Weighting

Step	Formula	Purpose
Step 1 Normalise	$r(i,j) = \frac{x(i,j)}{\sqrt{\sum_i x(i,j)^2}}$	Applied to each criterion before weighting
Step 2 Weight via AHP	$v(i,j) = w(j) \times r(i,j)$	AHP-derived weights across the four criteria
Steps 3 and 4 Distance to ideal	$D+(i) = \sqrt{\sum_j (v(i,j) - A+(j))^2}$ $D-(i) = \sqrt{\sum_j (v(i,j) - A-(j))^2}$	A+ is the best value per criterion and A- the worst. Each carpark is scored on its distance to both
Final score	$\text{PriorityIndex}(i) = \frac{D-(i)}{(D+(i) + D-(i))}$	Combines both distances into one rankable score per carpark
Sensitivity analysis	Perturb each weight by 10 to 20%, one-at-a-time or Monte Carlo, then re-rank	Robustness is measured by Spearman rank correlation and top-100 overlap against the baseline ranking

5. Scope of Work

5.1 Data Sources & Preparation

Dataset	Role
HDB Carpark Information	Base entity table: 2,272 carparks, location, type
LTA Electric Vehicle Charging Points	Existing charger supply, coverage gaps
HDB Property Information	Block-level commercial / market_hawker flags (Commercial Density)
Carpark Availability API	Occupancy proxy; car-lot counts (lot_type_code = C)
LTA Registration & Fuel-Type Series	National EV rate (no geographic breakdown)
HDB Resale Flat Prices (Jan 2017 onward)	Monthly house-value proxy by town (Section 4)
Resident Population by Planning Area/Subzone (Census 2020)	Subzone population share
URA Master Plan Planning Area/Subzone Boundary	Spatial joins (point-in-polygon)

Preparation: Geocode HDB Property Information and convert carpark coordinates from SVY21 to WGS84. Build 300 to 500m catchment buffers and point-in-polygon joins to planning-area and subzone boundaries. Filter Carpark Availability lots to car type only. Profile completeness and coverage against the 2,272-carpark base (Section 5.2).

5.2 Data Dictionary & Feasibility Statistics

Field	Data Type	Description	Source
carpark_id	String	Unique ID for each of 2,272 carparks	HDB Carpark Information
latitude / longitude	Float	Geocoded coordinates	HDB Carpark Information
total_lots	Integer	Car-lot count (lot_type_code = C)	Carpark Availability API
planning_area / subzone	String	URA planning area / subzone	URA Master Plan Boundary
commercial_density_score	Float	Weighted count of commercial / market_hawker blocks within 500m	HDB Property Information
ev_adoption_proxy_score	Float	4-step disaggregated EV adoption estimate	LTA Fuel-Type Series, HDB Resale Prices, SingStat Subzone Population, Carpark Availability API
charging_urgency_score	Float	Town-level EVCP occupancy disaggregated by lot share	Carpark / EVCP Availability data
evcp_saturation_score	Float	Existing EVCPs as a share of car-lot capacity	LTA EVCP, Carpark Availability API
priority_index_score	Float	Final TOPSIS-derived score	Derived
segment_label	String	Cluster / typology assignment	Derived

Feasibility statistics: Base population is confirmed at 2,272 HDB carparks. Phase 1 adds coverage and missingness per source, distribution checks on dwelling and commercial block counts and lot counts, resale transaction volume by town and quarter to flag which planning areas need the Section 4 fallback, and the share of subzones needing the population fallback.

5.3 Proposed Models, Techniques & Evaluation

Model	Purpose	Evaluation
MCDA: AHP + TOPSIS	Primary Priority Index ranking all carparks	AHP Consistency Ratio below 0.10; 10 to 20% weight sensitivity; backtest against known high-utilisation towns
K-means / hierarchical clustering	Demand-archetype segmentation	Silhouette score; Davies-Bouldin index; stakeholder review
Linear regression (stretch)	Sanity-check and recalibrate MCDA weights	RMSE / MAE, adjusted R-squared under spatial block CV; coefficient against AHP-weight comparison
GenAI / LLM layer	Insight-to-impact narrative generation	Human review for factual grounding before external use

5.4 Key Risks & Mitigations

Risk	Mitigation
EV-Adoption Proxy relies on house-value and population disaggregation assumptions. Tengah has zero recorded resale transactions (confirmed) and disclosure-suppressed 2020 Census figures	State assumptions explicitly; use a lease-filtered, multi-town resale proxy for Tengah, about 1.13 times the national median (Section 4); backtest and run sensitivity analysis
Car-lot counts available only for 2,003 of 2,272 carparks (88%). Commercial Density needs address-based geocoding of HDB blocks	Document imputation and exclusion rules; use a reliable batch geocoder; profile match rates in Phase 1
Carpark Availability data is a single snapshot, with staleness varying up to about ten months across records	Live polling over Phase 1; flag stale records rather than treating all as current
No grid or substation capacity data, and no capex or tariff data, exists	Use the density-based congestion proxy and utilisation-driven prioritisation; state as a limitation (Section 5.5)
GenAI-generated narratives could misstate the underlying model	Mandatory human review before any external stakeholder distribution

5.5 Out-of-Scope, Limitations & Assumptions

Out of scope and stated limitations: No grid or substation capacity data exists at carpark level, so a density-based proxy is used instead. No CapEx, tariff or installation-cost data exists, so the payback objective is addressed directionally through utilisation-driven prioritisation rather than an explicit financial model. The Carpark Availability API measures general lot occupancy, not EV-specific charging sessions.

Key assumptions: The 2,272-carpark scope reflects the current HDB carpark population at project start. Stakeholders are available for the AHP workshop within the Section 7 timeline. HDB towns and URA planning areas are treated as equivalent for the resale-price join (Section 4). Carpark Availability API access remains stable for the sampling window.

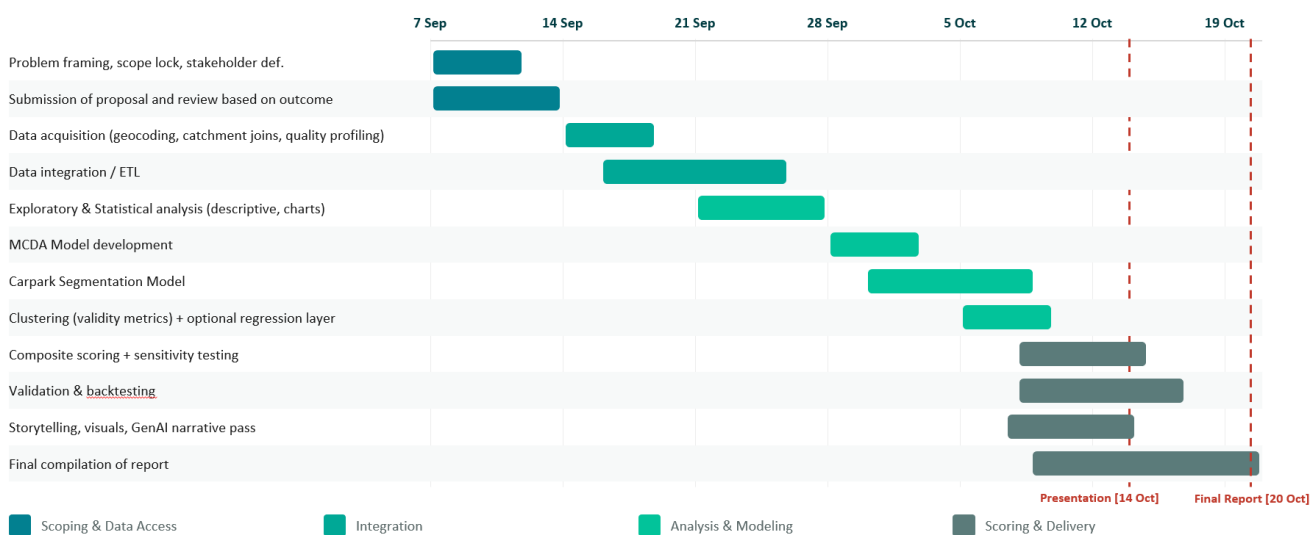
6. Key Deliverables

#	Deliverable	Description
D1	Integrated Geospatial Database	Carpark-level dataset (2,272 carparks) merging all 8 sources, with data dictionary
D2	EV Site Priority Index	MCDA (AHP + TOPSIS) ranked priority score per carpark, with weights and sensitivity analysis
D3	Carpark Segmentation Model	Clustering model and demand-typology report
D4	Interactive Dashboard	AI-augmented visualisation with GenAI-generated narrative summaries
D5	Data Management Plan	Data modelling, integration, quality, and security / privacy documentation
D6	Final Report & Presentation	Consolidated methodology, findings, validation, and recommendations

7. Effort Estimates and Timeline

Submitted 11 September 2026, due 20 October 2026. Six weeks, about 40 person-days, with several phases run concurrently to meet the deadline.

Gantt Chart [7 Sep to 20 Oct 2026]



8. References

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- [5] Ministry of Transport. "Utilisation Patterns of EV Chargers at HDB Car Parks." Parliamentary reply, 5 May 2026.
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